The Role of Generative AI Models in Requirements Engineering: A Systematic Literature Review

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Abstract

The software engineering field is experiencing rapid growth, driven by recent advancements in Artificial Intelligence (AI), particularly in Generative AI (GenAI) and Large Language Models (LLMs). Requirements Engineering (RE), a critical phase in software development, involves gathering and defining software requirements. However, research on the impact of GenAI and LLMs within RE remains limited. This paper examines the adoption of GenAI in RE, with the aim of exploring its practical implications, identifying current research trends, and highlighting areas for future development. To achieve this, a systematic literature review was conducted, addressing three research questions and analyzing 44 studies published over the past decade. The findings reveal that GenAI models, especially LLMs, are extensively employed in a variety of RE tasks, underscoring the versatility and potential of LLMs in enhancing the RE process.

CCS Concepts

• **Software and its engineering** → **Requirements analysis**.

Keywords

Requirements Engineering, Generative AI, Large Language Models

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1 Introduction

Generative AI (GenAI) is a subset of AI that can generate various types of content including but not limited to text, voice, video, music, and images. There are many types of GenAI models such as Generative Adversarial Networks (GAN), Transformer models, Large Language Models (LLMs), and Variational Auto Encoders (VAEs). Transformer models are a class of deep learning models specifically designed for natural language processing (NLP) tasks. The transformer architecture uses a self-attention mechanism that allows the model to weigh the importance of different words in a sentence when processing it [47]. Common examples of transformer models include Bidirectional Encoder Representations from Transformers (BERT) [17], Generative Pre-trained Transformer (GPT), Text-To-Text Transfer Transformer (T5), and Robustly optimized BERT approach (RoBERTa). LLMs are a type of GenAI and they are trained on large amounts of text data as well as utilize deep learning to generate human-like text. These models are pre-trained on an enormous amount of text data and can be fine-tuned for specific tasks by further training them on smaller, task-specific datasets. Common examples include OpenAI's GPT and BERT. In regards to natural language, their applications extend to various use cases such as text generation, language translation, content summarization, and code generation [32].

Requirements Engineering (RE) is an important phase in software engineering that involves the elicitation, specification, analysis, validation, and management of software requirements [27]. The primary goal of RE is to capture the needs of a stakeholder and ensure that their desired objectives are reflected in the final software product [46]. Natural language is widely used in every phase of the RE. It is a flexible medium for expressing, documenting, analyzing, and communicating requirements in RE. This provides a huge opportunity where GenAI can be applied to optimize the RE processes. By employing GenAI models, RE can be enhanced significantly in many ways. These technologies can be leveraged to interpret and process natural language data, offering capabilities such as semantic analysis, context understanding, language translation, and sentiment recognition.

This paper aims to explore the role of GenAI models in RE tasks. By analyzing the existing literature, this paper aims to provide an overview of the challenges, applications, and current research trends in this area. Research papers published between 2013 and 2024 have been reviewed as part of this study to understand how these technologies helped enhance RE. The final findings are summarized, analyzed for evidence, and presented in this paper.

2 Methodology

2.1 Research Questions

This study aims to answer the following research questions:

RQ1. Are there any GenAI models available that aid in RE activities?

RQ1.1. What types of requirements (textual use cases) artifacts do these models handle?

RQ1.2. Which RE activities do these models support?

RQ1.3. What approaches do these models follow for RE?

RQ2. How much evidence is available to utilize GenAI models in RE?

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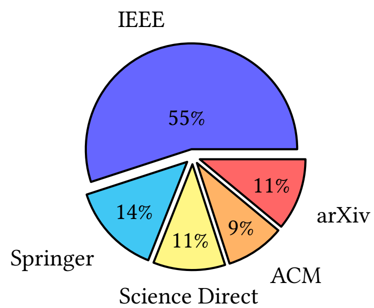

Figure 1: Composition of Literature Sources

RQ3. What are the limitations of current applications of GenAI models in RE?
Addressing RQ1 enables the exploration of the availability of GenAI models and their potential applications in RE activities. It also facilitates the identification of the models, an understanding of their functionalities, and an evaluation of their accessibility to practitioners. To ensure a comprehensive analysis and practical relevance of these models, RQ1 is further refined into sub-questions.
RQ1.1 delves into understanding the specific types or formats of requirements documentation and artifacts that these models can process. This investigation provides insights into the potential applications in RE by identifying what these models can handle, including but not limited to tasks such as text generation, summarization, translation, and classification.
RQ1.2 discusses the specific activities such as elicitation, analysis, specification, validation, and management supported by these models. It guides in understanding the model's applicability across different phases of RE.
RQ1.3 helps understand the approaches of GenAI models in RE tasks. This is crucial for making informed decisions about their adoption, customization, and optimization to meet specific organizational and project requirements.
RQ2 addresses the evidence supporting the adoption of GenAI models for RE. By evaluating the level of empirical evidence, ranging from demonstration examples to industrial practice, researchers and practitioners can measure the reliability of these methods.
RQ3 aims to identify and analyze the limitations inherent in current GenAI models. Understanding these limitations is crucial for mitigating risks, making informed decisions, and guiding future research efforts to enhance the effectiveness and applicability of these technologies in RE processes.

2.2 Search Process

This study aims to compile research papers on the applications of GenAI in Requirements Engineering (RE). The research papers published between 2013 and 2024 were included in the search. Both manual and automated search methods were employed as part of the search process. A search string was formulated using keywords specific to the study, such as "Requirements Engineering", "Generative AI," "Large Language Models", and "Artificial Intelligence". These keywords were divided into two main categories: RE-related keywords and AI-related keywords, connected by "AND" operator. Within each category, the keywords were separated by "OR" opeartor. The following search string was used:

(Requirement **OR** Requirements **OR** Requirements Engineering **OR** RE **OR** Software Requirements) **AND** (LLM **OR** Language Models **OR** AI **OR** Artificial Intelligence **OR** Generative AI **OR** GenAI **OR** Large Language Models)

The initial phase is the automatic search process, which spanned several databases including the ACM Digital Library, IEEE Xplore, Science Direct Elsevier, and Springer. The search was configured to match the search string in the full text and the metadata of the papers. The top 100 relevant papers were selected from each database which resulted in a total of 400 papers across all the databases combined. On the contrary, a manual search was conducted in Google Scholar web search engine. The top 100 results were analyzed and 16 relevant papers were chosen from this manual search process. For the next phase, the papers from the automatic search process were further refined by using the search string, specifically in the document title and abstract. Any possible duplicates were also eliminated in this phase. This resulted in a total of 58 papers. The next step was assessing the abstracts manually, removing duplicate studies, and considering the inclusion and exclusion criteria described in Section 2.3. The search and selection process of the research papers discussed above is illustrated in Figure 2. Finally, after following the process, a total of 44 research papers were included as candidates in this study. The resultant papers are shown in Table 1. The composition of the literature sources is illustrated in Figure 1, about 55% of the papers were from IEEE, 14% of the papers were found in Springer, followed by 11% and 9% in Science Direct and ACM respectively. During the manual search process, about 11% of related literature was found in arXiv database.

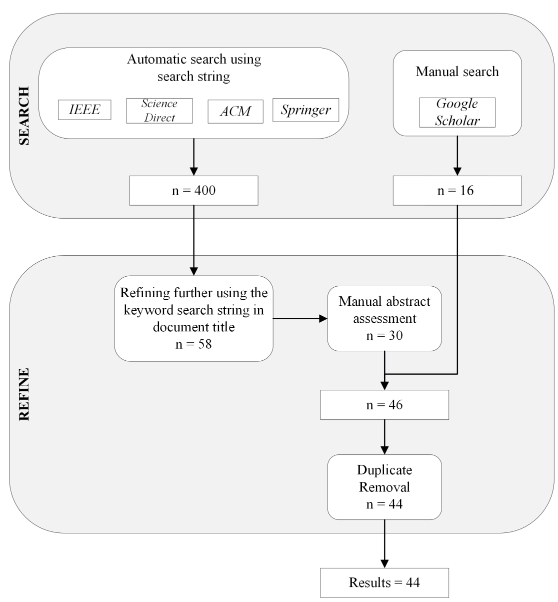

Figure 2: Process of Literature Search and Selection

Table 1: Selected Primary Literature Used in this Study

ID	Title	Source	Search Process	
P1	Generating Requirements Elicitation Interview Scripts with Large Language Models	IEEE	Automatic	[23]
P2	Using ChatGPT to Generate Human-Value User Stories as Creativity Triggers	IEEE	Automatic	[35]
P3	Improving Requirements Classification Models Based on Explainable Requirements Concerns	IEEE	Automatic	[24]
P4	Tool For Software Requirement Allocation Using Artificial Intelligence Planning	IEEE	Automatic	[38]
P5	Extracting Domain Models from Textual Requirements in the Era of Large Language Models	IEEE	Automatic	[6]
P6	A Transformer-based Approach for Abstractive Summarization of Requirements from Obligations in Software Engineering Contracts	IEEE	Automatic	[28]
P7	MAPE-K Loop-Based Goal Model Generation Using Generative AI	IEEE	Automatic	[36]
P8	On the Use of ChatGPT to Support Requirements Engineering Teaching and Learning Process	Springer	Automatic	[10]
P9	Investigating ChatGPT's Potential to Assist in Requirements Elicitation Processes	IEEE	Automatic	[40]
P10	Requirements Modeling Aided by ChatGPT: An Experience in Embedded Systems	IEEE	Automatic	[42]
P11	Large Language Model Assisted Software Engineering: Prospects, Challenges, and a Case Study	Springer	Automatic	[7]
P12	Inconsistency Detection in Natural Language Requirements using ChatGPT: a Preliminary Evaluation	IEEE	Automatic	[20]
P13	On the Use of GPT-4 for Creating Goal Models: An Exploratory Study	IEEE	Automatic	[11]
P14	Retraining a BERT Model for Transfer Learning in Requirements Engineering: A Preliminary Study	IEEE	Automatic	[2]
P15	BERT-Based Approach for Greening Software Requirements Engineering Through Non-Functional Requirements	IEEE	Automatic	[44]
P16	RE-BERT: automatic extraction of software requirements from app reviews using BERT language model	ACM	Automatic	[14]
P17	Improving BERT model for Requirements Classification by Bidirectional LSTM-CNN Deep Model	Science Direct	Automatic	[30]
P18	BERT-CNN: Improving BERT for Requirements Classification using CNN	Science Direct	Automatic	[29]
P19	PRCBERT: Prompt Learning for Requirement Classification using BERT-based Pretrained Language Models	ACM	Automatic	[34]
P20	Is BERT the New Silver Bullet? - An Empirical Investigation of Requirements Dependency Classification	IEEE	Automatic	[16]
P21	Extracting and Classifying Requirements from Software Engineering Contracts	IEEE	Automatic	[43]
P22	NoRBERT: Transfer Learning for Requirements Classification	IEEE	Automatic	[26]
P23	Large Language Models for Software Engineering: Survey and Open Problems	IEEE	Automatic	[19]
P24	GPT2SP: A Transformer-Based Agile Story Point Estimation Approach	IEEE	Automatic	[21]
P25	Extracting Goal Models from Natural Language Requirement Specifications	Science Direct	Automatic	[13]
P26	Zero-shot Larning for Requirements Classification: An Exploratory Study	Science Direct	Automatic	[4]
P27	Requirements Engineering and Large Language Models: Insights From a Panel	IEEE	Automatic	[9]
P28	Automatic Detection Method for Software Requirements Text with Language Processing Model	IEEE	Automatic	[50]
P29	Improving Requirements Completeness: Automated Assistance through Large Language Models	Springer	Automatic	[33]
P30	Advancing Requirements Engineering through Generative AI: Assessing the Role of LLMs	arXiv	Manual	[5]
P31	Which AI Technique Is Better to Classify Requirements? An Experiment with SVM, LSTM, and ChatGPT	arXiv	Manual	[18]
P32	Generative Artificial Intelligence for Software Engineering – A Research Agenda	arXiv	Manual	[37]
P33	ChatGPT Prompt Patterns for Improving Code Quality, Refactoring, Requirements Elicitation, and Software Design	arXiv	Manual	[49]
P34	Requirements Engineering using Generative AI: Prompts and Prompting Patterns	arXiv	Manual	[41]
P35	Accelerating Software Development Using Generative AI: ChatGPT Case Study	ACM	Manual	[39]
P36	ECHO: An Approach to Enhance Use Case Quality Exploiting Large Language Models	IEEE	Manual	[15]
P37	Comparing the Performance of GPT-3 with BERT for Decision Requirements Modeling	Springer	Manual	[22]
P38	Neural Language Models and Few Shot Learning for Systematic Requirements Processing in MDSE	ACM	Manual	[8]
P39	On the Relationship Between Similar Requirements and Similar Software: A case Study in the Railway Domain	Springer	Manual	[1]
P40	Extracting Functional Requirements from Design Documentation using Machine Learning	Science Direct	Manual	[3]
P41	Sentence Embedding Models for Similarity Detection of Software Requirements	Springer	Manual	[12]
P42	A Named Entity Recognition Based Approach for Privacy Requirements Engineering	IEEE	Manual	[25]
P43	Enhancing Requirements Elicitation through App Stores Mining: Health Monitoring App Case Study	IEEE	Manual	[48]
P44	She Elicits Requirements and He Tests: Software Engineering Gender Bias in Large Language Models	IEEE	Automatic	[45]

2.3 Inclusion and Exclusion Criteria

The relevant research papers with publishing dates ranging from January 1st, 2013 to January 31st, 2024 were identified in this study. The papers were included in our review based on the following criteria:

- GenAI for Software Requirements Engineering.
- AI for Requirements Engineering.
- LLMs for Requirements Engineering.

The papers were excluded from our review according to the following criteria:

- Research not related to RE tasks.
- Research that does not use GenAI models or approaches for RE.
- Editorials, workshops, keynote summaries, tutorials, and short studies.
- Research papers reporting the same study or duplicated work.

The selection decisions were made by two independent reviewers who followed the inclusion and exclusion criteria outlined in the paper. To ensure reliability and consistency, two reviewers assessed each study independently. Any discrepancies between the reviewers were resolved through discussion and consensus. The agreement rate between the reviewers was 95%, which reflects the robustness of the selection process.

3 Results and Analysis

Following the comprehensive search process outlined in Section 2, each article was reviewed to address the research questions proposed in Section 2.1. The subsequent sections will present and summarize the findings.

RQ1. Are there any GenAI models available that aid in RE activities?
Upon examining each research paper individually, the models employed in RE activities are identified. LLMs are the predominant models observed with applications in RE tasks, based on the accumulated research. Table 2 displays the list of LLMs used in the primary studies listed in this paper. The commonly employed language model is BERT, which is a deep-learning NLP model based on the transformer architecture and can be used for a variety of language tasks. As shownn in Figure 3, BERT is primarily used in about 45% of the studies. GPT is also largely utilized for RE tasks such as classification of requirements, analyzing requirements, extracting, and summarizing requirements. About 38% of GPT and other models in its foundational series such as GPT-2, GPT-3, GPT-3.5 turbo, and GPT-4 are employed for experiments in the studies. Other models, such as BARD, BART, T5, Pegasus, and others, account for approximately 5% of the models.

RQ1.1. What types of requirement artifacts do these models handle?
The LLMs identified in RQ1 are proficient in handling various types of textual requirement artifacts, given that they are pre-trained on language data and skilled at NLP tasks. In the reviewed literature, these models handle RE tasks such as classification, extraction, analysis, and validation of requirements sourced from artifacts such as documents, user stories, use cases, obligation statements, elicitation interview scripts, and even product application reviews.

Table 2: Large Language Models (LLMs) Utilized in the Literature and Their Providers

Providers	LLMs
OpenAI	GPT-2
OpenAI	GPT-3
OpenAI	GPT-3.5
OpenAI	GPT-4
OpenAI	Code-cushman:002
Google	BERT
Facebook	RoBERTa
Google	BARD
Facebook	BART
Google	T5
Google	Pegasus
Nvidia	Starcoder
Microsoft	All-MiniLM v2
HuggingFace	Sentence-BERT
EleutherAI	GPT-J-6B
DeepL SE	DeepL

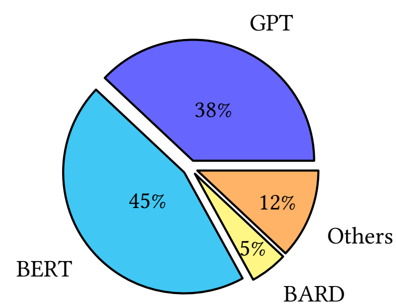

Figure 3: Composition of LLMs

Table 3: Evidence Assessment Metrics

Meaning	Strength
No Evidence	Weakest
Evidence from Demonstrations	Weak
Evidence from Expert Opinion or Observation	Normal
Evidence from Academic Studies	Normal
Evidence from Industrial Studies	Strong
Evidence from Industrial Practice	Strongest

RQ1.2. Which RE activities do these models support?
Based on the studies, the models support all major RE activities such as requirement elicitation, analysis, specification, validation, and management. The models are predominantly used in requirement analysis, closely followed by specification and elicitation. They were least applied in validation and management of requirements as shown in Figure 4.

RQ1.3. What approaches do these models follow for RE?
Prompt engineering, prompt patterns, retraining, and fine-tuning of the pre-trained and transformer-based generative models utilizing

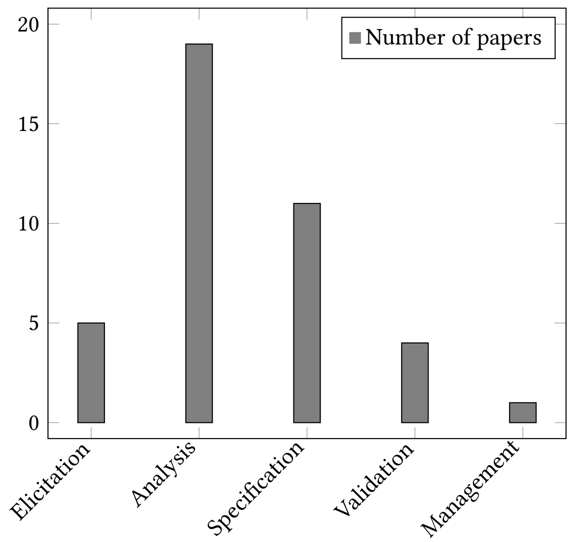

Figure 4: Requirement Engineering Activities Supported By LLMs

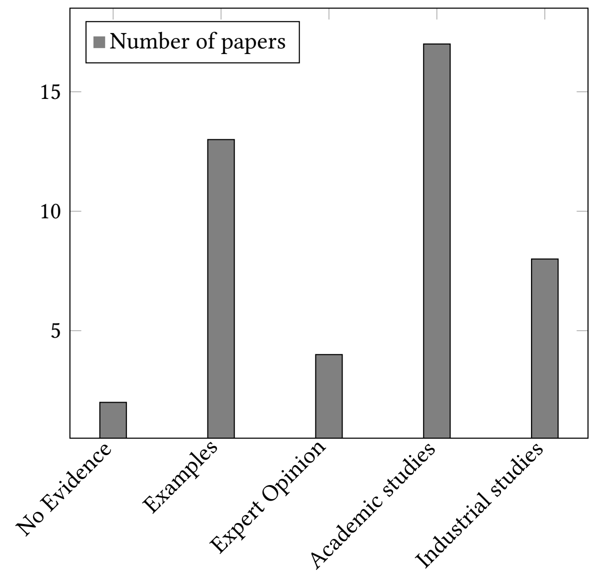

Figure 5: Literature Evidence Evaluation

NLP techniques, zero-shot learning, and transfer learning are some of the common approaches employed in the reviewed studies.

RQ2. How much evidence is available to utilize GenAI models in RE?
This study uses the evidence assessment criteria outlined in Table 3, based on Kitchenham's classification [31], to evaluate the strength of the evidence.

A significant portion of the literature, about 39%, relies on academic studies to evaluate and experiment with the proposed approach. These research methodologies include conducting experiments to collect empirical evidence or perform comparative analysis. About 30% of the studies utilize demonstrations with the help of prompts and prompt engineering. A notable portion of the literature, about 18%, draws evidence from industrial studies such as case studies addressing industry-specific challenges and constraints. Only 9% of the studies incorporate evidence derived from expert opinion, such as domain experts or practitioners in RE, and observations. As shown in Figure 5, only two papers were without associated evidence, as they were systematic literature reviews.

RQ3. What are the limitations of current applications of GenAI models in RE?
Even though there are various GenAI models and tools available, most of these studies utilize predominantly LLMs to conduct RE activities. Relying only on LLMs limits the exploration of other GenAI models that could offer better capabilities or address specific RE challenges more effectively. LLMs depend heavily on quality training data, inadequate or inaccurate training data might lead to biased outputs or sub-optimal performances. The inherent black-box nature of LLMs, which does not explain or interpret the inner workings of the model may lead to challenges in transparency and accountability.

4 Discussion

The main findings and limitations of this study are discussed in this section:

4.1 Findings

- *Prompt engineering*:
  Prompt engineering and prompt patterns are used widely in the majority of the studies [6, 18, 23, 28, 34, 35, 39, 49]. Some of the tasks where they are applied are as follows: generate requirements, generate elicitation interview scripts, summarize requirements, requirement classification, and extraction of requirements from textual documents.
- *Model training*:
  Techniques such as zero-shot learning [4, 42], transfer learning, and fine-tuning models are used in some studies [2, 3, 14, 18, 25, 29, 30, 44, 48]. By leveraging these fine-tuned models, requirements can be efficiently identified, classified, assessed for similarity, and extracted from documents or any other textual sources. BERT is fine-tuned on RE-specific datasets such as PROMISE and PURE datasets [12, 24] for aiding in tasks like requirements classification. Apart from model training and fine-tuning, NLP techniques such as Tokenizing, POS tagging, Sentence splitting, and lemmatization are utilized for efficient training of the language models [33, 50].
- *Other GenAI techniques:*
  MAPE-K (Monitor, Analyze, Plan, and Execute) Loop is a technique to control a software system by executing four steps in a loop. LLMs are utilized in the MAPE-K process to generate non-functional requirement goal models [36]. DeepL language translator was used in a study to identify

gender bias in RE, by analyzing pronouns in tasks. The tasks were translated to Finnish, a genderless language, and back to English using DeepL [45].

- *Goal modeling:*
  Goal modeling is closely related to RE and it helps describe the goals and capture the goals and objectives of a system. Studies [11, 13] discuss goal model generation from natural language requirement specifications.

4.2 Limitations

A significant limitation of this study is the lack of a variety of GenAI models and techniques explored for RE tasks. All of the examined papers can be classified into two main categories: employing AI methodologies to train transformers like BERT and use them for RE tasks, or utilizing LLMs like GPT and integrating them into RE tasks through prompt engineering. There is a possibility that there might be more effective GenAI tools and techniques that were not found during the search process for this study or they may not be developed or published yet. Furthermore, the search strategy in this study involved the use of a specific search string which may have not found other relevant papers beyond those included in this study. The study focused solely on the papers listed in Table 1 and did not examine the references cited within those papers, which may have resulted in the omission of additional information that could contribute to this research.

5 Conclusion

In this research, we conducted a systematic literature review aimed to identifying studies that explore the application of GenAI in RE. The study analyzed the selected papers to uncover the patterns, trends, and common practices in the utilization of GenAI in various stages of RE. The results indicate that GenAI models, specifically LLMs, are widely applied in various RE tasks, demonstrating the versatility of LLMs. Additionally, the existing studies provide insight into specific techniques such as prompt engineering and transformer-based model training. Although this review predominantly focuses on LLMs, future research could investigate the applicability and availability of other GenAI models and methods in RE tasks.

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